Mekong Lens

Interactive Water-Risk Visualization for the Vietnamese Mekong Delta

Project 2 Proposal — Data Stories with Python Shiny

Team Members: Tran Anh Chuong, Duong Hien Chi Kien, Nguyen Tran Nhat Minh, Phung Ba Quan

GitHub Repository: https://github.com/jimmytrn154/COMP4010-Project-2

1. Project Description

Mekong Lens is an interactive Python Shiny dashboard for exploring water-related climate risk in the Vietnamese Mekong Delta. The dashboard focuses on province-level rainfall, rainfall anomaly, drought signals, flood or surface-water patterns, and exposure indicators such as population and cropland. Its goal is to help users identify where and when water risk appears most strongly, and how different provinces experience different types of risk across seasons and years.

2. Central Question

Which Mekong Delta provinces face the highest water-risk patterns, and how do rainfall, flood or surface-water extent, and drought signals change across space and time?

3. Motivation

The Vietnamese Mekong Delta is one of the country's most important agricultural and livelihood regions, but it is increasingly exposed to abnormal rainfall, flooding, drought, and salinity intrusion. These risks vary across provinces and seasons, making them difficult to understand through raw datasets or static charts. Mekong Lens turns satellite and geospatial data into an interactive visual story that supports regional comparison and water-risk exploration.

4. Dataset Description

The project uses public satellite and geospatial datasets collected mainly through Google Earth Engine. The planned sources include FAO GAUL administrative boundaries, CHIRPS Daily rainfall, JRC Global Surface Water or the Global Flood Database for flood/surface-water proxies, and WorldPop or ESA WorldCover for population and cropland exposure. The rainfall analysis is planned for 2000–2024, with 2025 treated as an optional latest-year preview if complete enough.

The data pipeline filters the 13 Mekong Delta provinces, clips raster datasets to this region, aggregates daily rainfall into monthly province-level totals, and computes rainfall anomaly and drought-related indicators. Surface-water layers will be converted into monthly water-area indicators, while exposure layers will be aggregated into province-level population and cropland variables. The final structure is a province-month panel, where each row represents one province in one month.

The main limitations are spatial aggregation, mixed temporal coverage, and uncertainty in satellite-derived measurements. Surface water is also only a proxy for flood impact, so the dashboard will present risk scores as exploratory indicators rather than official disaster-warning outputs.

5. Visualization Challenge

This project is non-trivial because it combines spatial, temporal, and raster-based environmental data. A static chart cannot show how rainfall, drought, surface water, and exposure vary across both provinces and months. The dashboard therefore uses linked views: an interactive map, time-series charts, anomaly heatmaps, ranking charts, and comparison panels. Users can filter by year, month, province, and metric, then move from a regional overview to province-level detail.

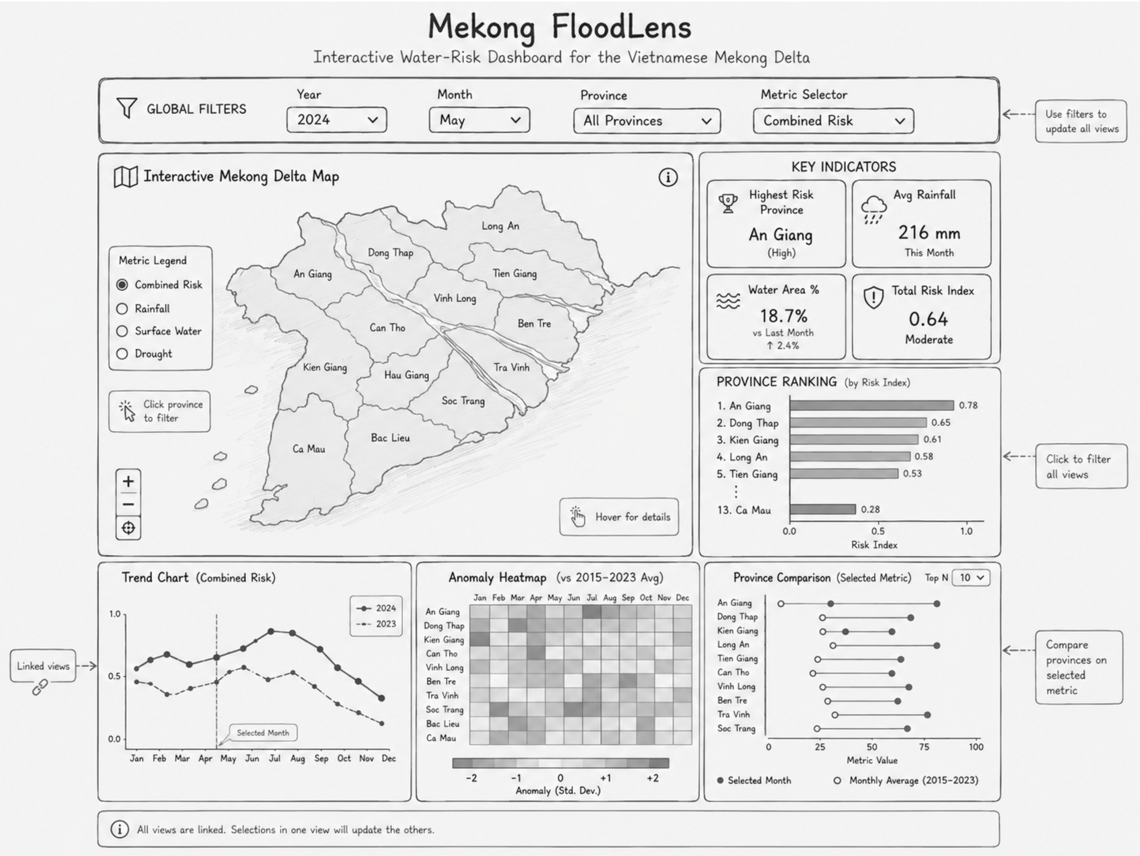

Figure 1: Initial dashboard wireframe for Mekong FloodLens.